Chemistry - Undergraduate Level Assessment 13 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Elements with partially filled 4 f or 5 f orbitals include all of the following EXCEPT
   (a) Cu
   (b) Gd
   (c) Eu
   (d) Am
2. The strongest base in liquid ammonia is
   (a) NH 3
   (b) NH 2-
   (c) NH 4+
   (d) N_2H_4
3. Reduction of D-xylose with NaBH 4 yields a product that is a
   (a) racemic mixture
   (b) single pure enantiomer
   (c) mixture of two diastereomers in equal amounts
   (d) meso compound
4. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
   (a) 23.56 GHz
   (b) 42.58 MHz
   (c) 74.34 kHz
   (d) 13.93 MHz
5. The fact that the infrared absorption frequency of deuterium chloride (DCl) is shifted from that of hydrogen chloride (HCl) is due to the differences in their
   (a) electron distribution
   (b) dipole moment
   (c) force constant
   (d) reduced mass

True / False (10 marks)

Answer True or False

6. True or False: Both paramagnetism and ferromagnetism require the presence of unpaired electrons in their atomic or molecular structures.
7. True or False: The molecular geometry of thionyl chloride, SOCl 2, is tetrahedral because it has four regions of electron density around the sulfur atom.
8. True or False: Most organic compounds do not contain silicon atoms, making tetramethylsilane an ideal reference for 1 H NMR spectroscopy.
9. True or False: Cr_3+ cannot be used as a paramagnetic quencher due to its specific electronic configuration.
10. True or False: Raman spectroscopy is a light-scattering technique used to analyze molecular vibrations and chemical compositions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the balancing of the redox reaction represented by the equation MnO 4- + I- + H+ <-> Mn_2+ + IO 3- + H_2O, and analyze the stoichiometric relationship between MnO 4- and I-.
12. Discuss the implications of the Heisenberg uncertainty principle for a quantum mechanical particle in the lowest energy level of a one-dimensional box, focusing on the limitations of measuring position and momentum.

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